

RAEYAAN MUPPANENI

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EDUCATION

Irvington High School Class of 2027

Fremont, California

- GPA: 4.71 weighted / 4.06 unweighted.
- Advanced coursework: Multivariable Calculus, Calculus II, Quantum Physics, AP Calculus AB, AP Physics C: Mechanics & E&M, AP Chemistry, AP Computer Science A, AP English Language, AP Research, AP Seminar, AP Environmental Science, AP Macroeconomics, AP U.S. Government, AP World History, Spanish 3.
- AP Physics C Teaching Assistant (Grade 12).
- Clubs: Math, Robotics, Cricket.

RESEARCH & ENGINEERING

Stanford Institutes of Medicine Summer Research Program (SIMR) — Bioengineering June 2026–Present

Stanford University / Human Performance Lab

- Developed a capacitive sensing insole for para-athletes to study loading and gait asymmetry; designed the rigid PCB and contributed to electronics integration, firmware, calibration, and testing.
- Continued the project after SIMR with Stanford Human Performance Lab mentorship; leading a five-student team through the next hardware iteration, code integration, calibration, and testing, while advancing the project for the Conrad Challenge.

EMG-Controlled Robotic Arm / Exoskeleton Research 2024–Present

Polygence / Independent Research

- Built an EMG-controlled robotic arm that translates muscle signals into mechanical movement; investigated sensing, signal processing, and actuation.
- Earned 2nd place at the Alameda science fair. Research paper prepared for publication/conference submission; final publication status to be verified before external use.

Assistive Technology & AI Projects 2023–Present

- Built a wearable using YOLOv8/COCO object detection to warn visually impaired users of nearby obstacles; earned 2nd place at the Synopsys science fair and continued improving the wearable.
- Developing a handheld communication device for neurodivergent/nonverbal children with simple customizable one-press messages.
- Built an EEG-based focus application for neurodivergent users; filed a patent application for a medical-emergency wearable and has worked on related research documentation.

Applied Mathematics & Robotics Research 2024–Present

- Studied multivariable calculus and Jacobians in connection with robotic motion; explored group theory and open problems while documenting notes, proofs, and exercises in LaTeX.

LEADERSHIP & COMMUNITY

Rooting Minds — Co-Founder & President 2025–Present

Neurodivergent youth education & assistive technology

- Co-founded an initiative creating accessible games, workshops, and technology for neurodivergent children; partnered with FCSN and WeEMBRACE.
- Designed and distributed educational card decks, led workshops and teammates, and reached approximately 100 students; redesigned sessions around student participation and feedback.

FCSN & WeEMBRACE — Volunteer 2024–Present

- Teach and support neurodivergent children through badminton/athletics, robotics, coding, drawing games, and weekly educational activities.

Youth Council — Chapter President 2024–Present

- Planned agendas and monthly activities, coordinated officer responsibilities and communications, taught math enrichment to 38 younger students, and organized STEM/robotics and service activities.

INTERNSHIP

Fluora Bioscience Laboratories — Electronics Engineer Trainee Grade 12

7 hr/wk, 6 wk/yr | Will continue

- Electronics trainee on a portable fluorometer for noninvasive diabetes screening; designed LED-drive, photodetector-amplifier, and battery circuits.

MATHEMATICS & HONORS

Mathematics Competitor 2023–Present

Rigorous Mathematics / AMMOC

- USAMO Qualifier; AMC 12A: 139.5; AMC 12B: 144.
- Selected honors: High Distinction, Australian Mathematics Competition (2025); Distinction, Cayley Contest (2025); Honours, COMC (2024); AMO Silver and Bronze; Australian Mathematics Competition Distinction (2023); IYMC finalist; additional international math competition medals.

ELECTRICAL & ELECTRONICS SKILLS

- **PCB & Circuit Design:** KiCad schematic capture and multilayer PCB layout; 4-layer rigid and 2-layer flex PCBs; LC sensing circuits; LED-drive, photodetector-amplifier, and portable battery/power circuits.
- **Embedded Systems & Interfaces:** STM32 Nucleo WB55RG, TI FDC2214 capacitive-sensing IC, embedded firmware/code, sensor integration, I2C/SPI/UART interfaces, hardware/software integration, and data acquisition.
- **Sensors & Biomedical Instrumentation:** Capacitive sensing, EMG sensing and signal interpretation, photodetection/optical sensing, EEG-data applications, wearable sensors, gait instrumentation, and assistive/medical-device electronics.
- **Analog / Signal Engineering:** Amplification, filtering, LC oscillation/frequency measurement, driven shielding/active guarding, grounding, decoupling, noise reduction, signal-integrity considerations, calibration, and time-series signal processing.
- **Prototyping & Validation:** PCB bring-up, electronics debugging, component/datasheet selection, prototype integration, known-load calibration, experimental testing, iterative hardware redesign, and robotics actuator/sensor integration.

ROBOTICS & COMPUTING

Robotics 2023–Present

BISV competition robotics / Evodyne Robotics

- Engineer and driver on a four-student competition team; qualified for regionals and advanced to elimination rounds.

- Built a robotic dog and contributed to humanoid-robot development, including actuator design and cooling-system work.

Programming & AI 2022–Present

- Python, Java, and C; coursework/projects in algorithms, data structures, OOP, recursion, USACO Bronze, AP Computer Science A, and AI/ML.
- Completed Empowerly AI Scholars training and developed an image-classification project to identify distracted drivers.

ADDITIONAL ACTIVITIES

Cricket — Captain, American School of Cricketing Excellence 2019–2025

- Competed in 5+ tournaments and 50+ matches annually; trained 3+ times per week, captained teams, won multiple tournaments, and qualified for West Coast intrazonals.

Speech & Debate 2022–2023

- Competed in Lincoln-Douglas, Public Forum, Impromptu, and Oratorical Interpretation; placed 1st in Lincoln-Douglas at Golden State Academy and reached Public Forum elimination rounds.